

28. (2 points)

a. (0.75 point)

Identify and describe one “soft” approach to modeling the underwriting cycle.

b. (0.75 point)

Identify and describe one technical approach to modeling the underwriting cycle.

c. (0.5 point)

Briefly describe one feature that econometric modeling of the underwriting cycle has in common with “soft” approaches and one feature that it has in common with technical models.

END OF EXAMINATION

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 28	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE: C8: Describe approaches to modeling the underwriting cycle.
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answer 1</i> Delphi Method. Used to obtain expert consensus. Experts are given background information + a questionnaire. Responses are aggregated + presented to participants. Based on this participants can change their response or articulate the reasons for not agreeing. Process continues until consensus is achieved <i>Sample Answer 2</i> Competitor analysis – Combine information from trade publications, rate filings, agents, financial statements etc. to try to predict a turn in the UW cycle. The goal is to see if an unusually high number of competitors appear to be either financially distressed or very profitable <i>Sample Answer 3</i> Scenario testing: Written description of the future state of the insurance environment. Prepare management to think possible responses	
Part b: 0.75 point	
<i>Sample Answer 1</i> Autoregressive (AR(n)) Time Series. The industry combined ratio X_t is modeled as an autoregressive time series, generally $n=2$ or 3 autoregressive series work well. $X_t = a + \sum_{j=1}^n b_j X_{t-j} - j + \partial \in$ where $\in$ is standard normal variable. This is a mean reverting process with autocorrelation coefficient + an annual disturbance distribution. <i>Sample Answer 2</i> General factor model: $Z_t = a + b \times Z_{t-1} + \sigma \times \varepsilon_i$ $X_t = d + d \times (Z_{t-1} - X_{t-1}) + \tau \times \delta_i$	
Part c: 0.5 point	
<i>Sample Answers for a "soft" approach</i> - Both soft and econometric approach need large quantity, variety and complexity of data - Econometric modeling includes the recognition of human factors impacting the UW cycle making it similar to soft approach. - It is a mixture of both while incorporating structural insight of soft approach <i>Sample Answers for technical models</i> - Both technical + econometric models require mathematical formalism + rigor greater than soft approaches. - Technical => Statistical validity	
EXAMINER'S REPORT	
In general candidates did not do well on this item, especially on part c.	

EXAM 7 SPRING 2016 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part a
Candidates were expected to know one of the three soft approaches presented in the text and be able to provide a high-level description of it. The most common reason candidates lost credit was for an insufficient description of the chosen approach. Most candidates chose to discuss the Delphi method.
Part b
Candidates were expected to know one of the three technical approaches and to be able to provide a high-level description of the approach chosen. Formulas underlying a method were also accepted as characteristics. Note that in the second sample answer provided above, readings outside the syllabus incorporate σ into the ϵ term. This was accepted for full credit, and similarly for the second equation.
Part c
To receive full credit, candidates had to clearly identify the similarity and relate it to the econometric model. No credit was given for merely listing characteristics of econometric modeling without mentioning how it is similar to the other methods (soft or technical). One common reason candidates lost credit was for providing an insufficient description, such as “considers human behavior” or “judgment” for similarity with soft modeling.